Deep Reinforcement Learning-Enabled Anti-Reactive Jamming Strategies for Mobile Cell-Free Interference Networks

Tri Nhu Do[†] Georges Kaddoum[⋆] François Leduc-Primeau[†]

[†]Polytechnique Montréal

[⋆]École de technologie supérieure (ÉTS)

IEEE CCECE 2026

Introduction & Motivation

Mobile Cell-Free Interference (MCFI) Network:

- Multiple mobile transmitter–receiver pairs operating *concurrently*.
- Generalizes interference channels and cell-free network principles over time and frequency domains.
- Possible applications:
  - Tactical wireless networks (e.g., frequency hopping).

Reactive Jamming:

- Jammer stays silent until it detects active transmissions.
- Attacks the *strongest* active link (highest overheard SNR).

Challenges:

- Hub knows legitimate CSI, but jammer CSI is *not directly observable*.
- Attack is only detected *after* performance loss occurs.
- Need an adaptive, cognitive countermeasure.

Sec. II
Network & Signal Model

- MCFI network with N pairs, K bands, Hub, jammer
- Reactive jammer and jamming viewpoint

Sec. III-A
SE Maximization Problem

- Maximize sum spectral efficiency (SE)
- Collision-free frequency allocation (CFFA) constraints
- Formulated as a mixed-integer nonlinear programming problem

Sec. III-B
MDP Reformulation

- Reformulate problem as a Markov Decision Process (MDP)
- Handle the case where jammer's CSI is unknown
- State, action, reward designed for the CFFA setting

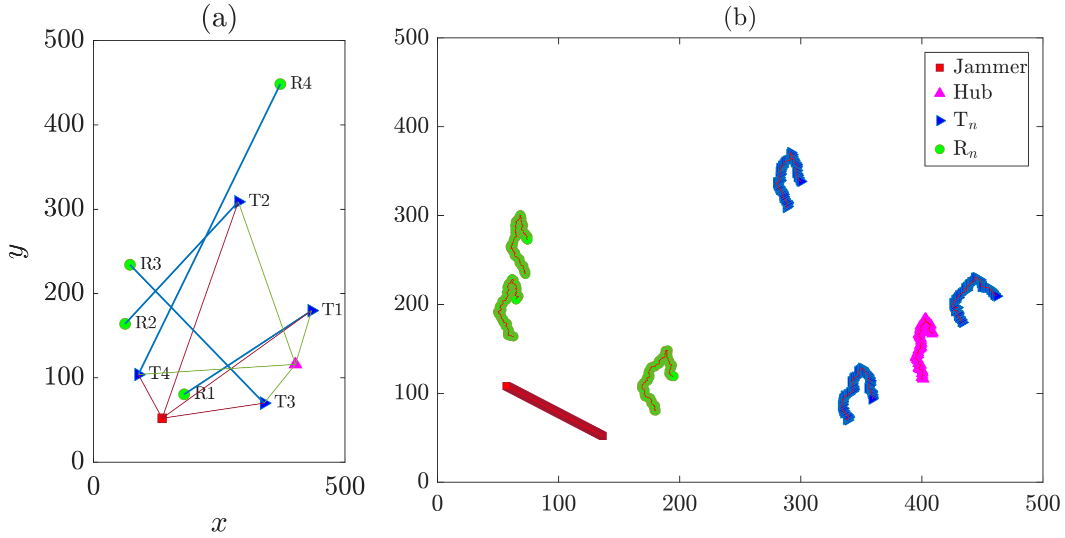
Sec. III-C
DRL Algorithm

- Model-free, off-policy deep reinforcement learning
- Deep Q-network (DQN) with experience replay buffer
- ϵ-greedy exploration with decay schedule

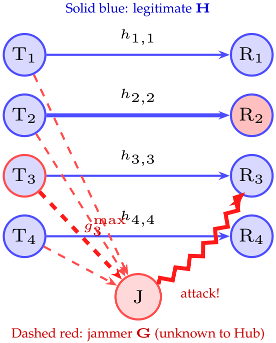
Sec. IV
Results & Validation

- Mobility models: reference point group mobility (RPGM) for users, random waypoint mobility (RWP) for jammer
- Convergence to a near-optimal policy
- Validate by exhaustive search

Network Description – MCFI snapshot & mobility patterns



Jamming Viewpoint



Jammer's view (G):

- Overhears all K bands.
- Picks $f_J^* = \arg\max_n \texttt{SNR}_{T_n J}^{f_{J,n}}$.

Hub's view (H only):

- Sees its own channel quality.
- Cannot observe **G**.
- *Different* ranking of strongest channel over different bands.

Consequence for the Learning Agent

The Hub only learns **which channel was attacked** via the missing ACK at time t_τ:

$$\mathbb{I}(f_{D,n}[t_\tau]) = 0 \;\Rightarrow\; \text{posterior evidence that } f_{D,n} = f_J^*$$

$\Rightarrow$ The RL algorithm must learn from *post-attack feedback only* $\Rightarrow$ always suffer performance loss

Spectral Efficiency Maximization

Network sum-SE under policy π_{t_τ}:

$$\mathrm{SE}_{\sum}[t_\tau] = \sum_{n=1}^{N} R^{f_{\mathrm{D},n}^{\pi_{t_\tau}}}[t_\tau]\,\mathbb{I}(f_{\mathrm{D},n}^{\pi_{t_\tau}}[t_\tau])$$

Optimization Problem:

$$\begin{aligned}
\max_{\boldsymbol{f}^{\pi_{t_\tau}}} \quad & \mathrm{SE}_{\sum} \\
\text{s.t.} \quad & f_{\mathrm{D},n}^{\pi_{t_\tau}} \neq f_{\mathrm{D},l}^{\pi_{t_\tau}}[t_\tau],\ \forall n, l, t_\tau \quad \text{(no collision)} \\
& |\mathcal{F}_{\mathrm{J}}[t_\tau]| = 1,\ \forall t_\tau \quad \text{(single jammed band)} \\
& R^{f_{\mathrm{D},n}^{\pi_{t_\tau}}}[t_\tau] = 0 \text{ if jammed}
\end{aligned}$$

Equivalent MDP Formulation

The problem is reformulated as a finite MDP $\langle \mathcal{S}, \mathcal{A}, p, r \rangle$:

State Space $\mathcal{S}$

$$\boldsymbol{s}_l = [s_{f_1}, \ldots, s_{f_K}, f_{\mathrm{J}}^*]$$

where $s_{f_k} \in \{\texttt{u1}, \ldots, \texttt{uN}, \varnothing\}$ is the in-use status; f_{J}^* inferred from *posterior* observations.

Action Space $\mathcal{A}$

$$\boldsymbol{a}_m = [f_{\mathrm{D},1}, \ldots, f_{\mathrm{D},N}]^T$$

Generated as *permutations without repetition* $\Rightarrow$ collision constraint enforced.

Reward

$$r(\boldsymbol{s}, \boldsymbol{a})[t_\tau] = \sum_{n=1}^{N} R_{f_{\mathrm{D},n}[t_\tau] | \boldsymbol{a}[t_\tau]}$$

Deep Q-Network (DQN) Approach

- **DQN**
  - Transition probability $p(\boldsymbol{s}', r \mid \boldsymbol{s}, \boldsymbol{a})$ is *unknown* (depends on $\mathbf{G}$).
  - Model-free, off-policy, online learning fits the continuing-task setting.
- **Observation vector:**

$$\boldsymbol{o}[t_\tau] = [s_{f_1}, \ldots, s_{f_K}, h^{f_1}_{\mathrm{T}_1,\mathrm{R}_1}, \ldots, h^{f_K}_{\mathrm{T}_N,\mathrm{R}_N}, f^*_{\mathrm{J}}]$$

- **Bellman update:**

$$Q(\boldsymbol{s}[t_\tau], \boldsymbol{a}[t_\tau]) \leftarrow Q(\boldsymbol{s}[t_\tau], \boldsymbol{a}[t_\tau]) + \alpha\Big(\underbrace{r[t_\tau] + \gamma \max_{\boldsymbol{a}'} Q(\boldsymbol{s}[t_{\tau+1}], \boldsymbol{a}')}_{\text{target}} - \underbrace{Q(\boldsymbol{s}[t_\tau], \boldsymbol{a}[t_\tau])}_{\text{predicted}}\Big)$$

Algorithm Components

- **ϵ-Greedy Action Selection (with decay):**

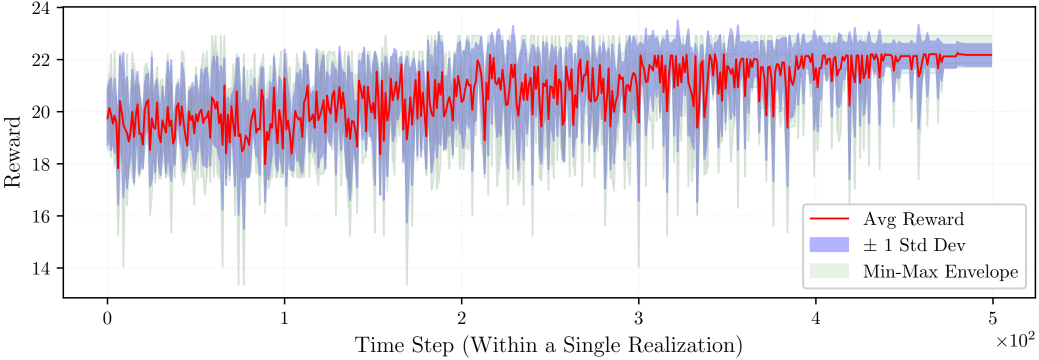
$$\epsilon[t_\tau] = \max\{\epsilon_{\texttt{start}} - \kappa[t_\tau]\,\epsilon_{\text{dec}},\ \epsilon_{\min}\}$$

  where $\epsilon_{\text{dec}} \triangleq 1/(\Lambda - \Lambda\beta\%)$; β = exploitation portion.

- **Experience Replay:** Store transitions $(\boldsymbol{o}, \boldsymbol{a}, r, \boldsymbol{o}')$ in buffer $\mathcal{B}$; sample mini-batches.
- **Two virtual Q-Networks:**
  - Q-network $Q(\boldsymbol{s}, \boldsymbol{a}; \boldsymbol{\theta})$ – predicted values.
  - Target Q-network $Q(\boldsymbol{s}, \boldsymbol{a}; \boldsymbol{\theta}^-)$ – stable targets.
- **Loss Function (MSE):**

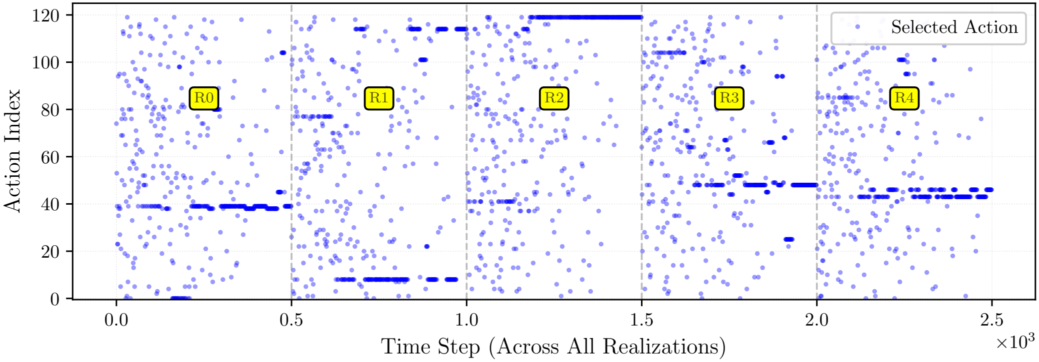
$$\texttt{loss}(\boldsymbol{\theta}[t_\tau]) = \mathbb{E}\Big[\big(r[t_\tau] + \gamma \max_{\boldsymbol{a}'} Q(\boldsymbol{s}[t_{\tau+1}], \boldsymbol{a}'; \boldsymbol{\theta}^-) - Q(\boldsymbol{s}[t_\tau], \boldsymbol{a}[t_\tau]; \boldsymbol{\theta})\big)^2\Big]$$

Numerical Results – Reward Convergence



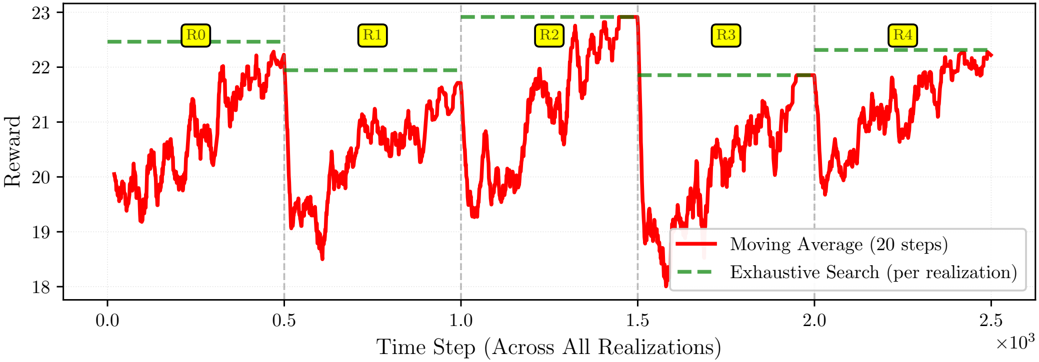
DRL-based algorithm converges to the value found by *exhaustive search*
$\Rightarrow$ validates correctness.

Numerical Results – Predicted Actions



Exploration phase (varied actions) → exploitation phase (best Q-value action).
Speed controlled by ϵ_{dec}.

Numerical Results – Performance Over Time Slots



SE across mobility realizations.

Conclusion

- Proposed DRL-based CFFA for jammed MCFI networks:
  - Maximizes *unjamming-effective* spectral efficiency.
- Propose the DRL algorithm and validate with exhaustive search.
- Numerical results:
  - Mobility (RPGM users, RWP jammer) affects SE.
- Future work: cognitive radio framework with real-time situational awareness.

Code available at

`https://github.com/TND-lab/DRL-Active-Jamming-Interference-Networks`

Thank you for your time and attention.